

fh6parse Linux kiosk manual

Operator sheet (Polish, daily use only): [LINUX-KIOSK-PL.md](#). Pi 5 pinout and wire poster: [LINUX-KIOSK-WIRING.pdf](#). Hardware buy list (SKU / EAN if shops die): [HARDWARE.md](#). This file is the full English install / update manual.

Version 1.4.0. Raspberry Pi 5 kiosk: portrait **800×600**, MUNBYN **P047**, USB /dev/usb/lp0 print, isometric **line-art** STEP (stick .stp first), **D** vs T warnings, programmed cycle time and a per-tool share chart. Two knobs: **file** (list) and **mill** (name next to **v1.4.0**). Three 16 mm vandal print buttons under the screen, left→right **green LOAD** (operator slip), **yellow SET** (setter), **red RUN** (full ticket). The same three colour chips sit on the bottom of the panel. List, preview, and those chips use large high-contrast type. An optional fourth GPIO button sleeps and wakes the panel (leave it unwired). An optional Pi 5 **J2** switch is the hardware power button (not GPIO). Highlight a file to preview ops, cycle time, 3D ready, and the stacked isometric before print. Add mills in **settings** (**Add mill...**: name, rapids m/min, B/C rapid, tool-change time, optional max rpm, optional G53 ATC X/Y/Z, work offset X/Y/Z, travel min/max per axis). Tickets then show a labeled G53 rectangle of where the work offset may sit, and max Ø for a centred outside G41/G42. Screen language is **Polish** by default (**F2 / C** / the **PL-EN** chip for English). A **wireframe 3D cube** next to a file means the STEP views are ready. Later git commits keep the **1.4.0** badge; the yellow **UPDATE** button then shows a short hash. The Windows / Linux office GUI is one report with a **section checklist** (no LOAD / SET / RUN presets). The same single-click yellow **UPDATE to ...** bar appears under the mill/print row: frozen Windows exe against a GitHub Release (tag **vX.Y.Z** builds it), git checkout against origin.

Python **3.10+** is required (Bookworm ships 3.11). Use **Raspberry Pi OS 64-bit Desktop** (Bookworm or later). Pi 5 has no 32-bit OS.

The 40-pin header uses the **same BCM numbers as Pi 3/4**. GPIO on Pi 5 goes through the **RP1** chip: **RP1.GPIO** does **not** work. The kiosk uses gpiozero with **lgpio**. Still switch the desktop to **X11** (tkinter + xset blanking).

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1. What you need

Item	Notes
Raspberry Pi 5	Official 27 W USB-C PSU (5 V / 5 A). Do not use a Pi 3 2.5 A supply. Do not power the printer from the Pi USB.

Item	Notes
Raspberry Pi Active Cooler	Required in the closed kiosk box. Heatsink + PWM fan on the SoC FAN header. Botland RPI-23925 . \$2.12 .
Pimoroni NVMe Base Duo (PIM704)	Under the Pi 5. PCIe FPC, not GPIO. Botland . Do not use the Raspberry Pi M.2 HAT+ from the SSD kit.
Official Raspberry Pi NVMe 512 GB (2230)	Disk only from Raspberry Pi SSD Kit 512 GB (RPI-25484). Unscrew it from the kit HAT+ and fit slot A on the Duo. Second Duo slot empty.
micro-HDMI cable	Pi 5 has two micro-HDMI ports. Use HDMI0 (the port next to USB-C power) for the kiosk panel. Clamp USB-C and HDMI0 to the case. No USB-C extension on the 27 W PSU.
Case (3D print)	PETG , not PLA. Vents above the Active Cooler and below the Duo. \$2.13 .
Spare PCIe flex (drawer)	Pimoroni PIM703 (35 mm) or PIM702 (50 mm). The OS cable; clips break.
800×600 LCD, mounted vertically	After rotation the framebuffer is 600×800 . That is what the app uses.
DFRobot Fermion EC11 (file)	SEN0235. Phase A and B only. Shaft push (C) unused. Selects the NC file.
DFRobot Fermion EC11 (mill)	Second knob, same wiring. Cycles mills in settings / kiosk.machine.
Three 16 mm 5-pin vandal buttons	Momentary, ring LED 5 V: green LOAD, yellow SET, red RUN . Mount under the screen , left→right matching the colour chips on the panel.
Spare momentary button	Optional. Same switch wiring as LOAD (NO → GPIO 25, C → common GND). Press blanks the panel; press again wakes. Leave unwired: the pin sits on the internal pull-up and never fires. Idle timeout, knobs, USB, and keyboard still sleep/wake.
Pi 5 J2 power switch	Optional. Momentary NO across the two J2 (PWR_BTN) pads — same as the PCB power button. Not a 40-pin GPIO. Omit it: USB-C still boots when you plug in.
USB stick	FAT/exFAT/NTFS. Programs as .nc / .NC / .tap in the stick root only (not subfolders).
Panel USB 3.0 Type-A (27 mm)	Metal socket on the enclosure for that stick. Allegro . Pigtail to a Pi USB 3 port. \$2.11 .
MUNBYN P047 (ITPP047)	USB, 80 mm ESC/POS, auto-cutter. Own mains PSU.
Company STEP folder (optional)	NAS of .stp / .step if the stick has none. See \$3.7 . Stick copy is enough.
Network (optional)	Only for git install and later UPDATE . Printing works offline.
Keyboard / mouse	First-time setup, settings (language, mill / Add mill... , file and mill knobs, RUN / LOAD / SET / optional sleep pins, encoder ticks), SSH, or tap UPDATE . Not required for encoder + GPIO print.

Default GPIO (**BCM** numbers, not header pin numbers):

Function	BCM GPIO	Header pin
File encoder A (CLK)	17	11
File encoder B (DT)	27	13
Mill encoder A (CLK)	5	29
Mill encoder B (DT)	6	31
RUN button (red)	22	15
LOAD button (green)	23	16
SET button (yellow)	24	18
Spare button (optional sleep/wake)	25	22

Function	BCM GPIO	Header pin
3.3 V for encoder VCC	—	1 or 17
5 V for button LED rings	—	2 or 4
GND	—	6, 9, 14, 20, or 30

Change pins in **settings** (**F2** / **C** / **PL-EN**) or in `/etc/fh6parse-kiosk.ini`. Settings write `~/.config/fh6parse/ui.ini` (and the main ini if it is writable).

Land every GPIO / 3.3 V / 5 V LED wire on the **screw terminal board** on the 40-pin header, or on a **GPIO riser** if the cooler is in the way (**\$2.9**). Do not also push Dupont onto the same header under that board.

2. Hardware

Wiring poster (print this at the bench): [LINUX-KIOSK-WIRING.pdf](#) — Raspberry Pi 5 top view, colour 40-pin map, panel layout, EC11 / vandal / J2, **screw terminal**, **NVMe Base Duo**, **panel USB**, **Active Cooler**, wire gauge and colours. Source: [LINUX-KIOSK-WIRING.html](#). The sections below are the same facts in text.

Buy list (survives dead shop links): [HARDWARE.md](#) — SKU / EAN / size / substitute, **must do** and shop suggestions. Manufacturer PDFs in [hardware-archive/](#). Assemble checklist: **\$2.13**.

2.1 Power and USB

- Pi 5 on the official USB-C 5 V / 5 A supply. USB-C on the Pi is **power only**. **Clamp** that plug and **HDMI0** to the case. Do **not** use a USB-C extension.
- Optional panel **power** is **J2**, not GPIO — **\$2.8**. Without it, plug in USB-C and the Pi boots.
- P047 on its own supply; USB cable to a Pi **USB 2** port is **data only** if the printer has a separate PSU.
- USB stick through the **27 mm panel USB 3.0** socket — **\$2.11**. Automount under `/media/<user>/...` or `/run/media/...` is enough; the kiosk polls those paths.
- Official **Active Cooler** on the SoC FAN header — **\$2.12**. Seat that cable before the GPIO screw terminal.
- OS disk is the official Raspberry Pi **512 GB NVMe** on the **NVMe Base Duo under the Pi** — **\$2.10**. Not GPIO. Not the kit M.2 HAT+.

2.2 GPIO rules

Pi GPIO is **3.3 V** (unchanged on Pi 5). Do not feed 5 V into A, B, or the button **signal** pins (C / NO / NC).

DFRobot EC11 (SEN0235) is **3.3–5 V**. Power it from the Pi: **VCC** → **3.3 V** (header pin 1), **GND** → **GND**. Do not use the Pi 5 V pins for the encoder. If a clone only runs at 5 V, you still must not put 5 V on the Pi inputs (use a level shifter).

Encoder **C** (shaft click): leave unconnected. The kiosk does not read the push-switch.

2.3 16 mm 5-pin vandal buttons (no datasheet)

The Allegro 5 V ring-LED parts (and the common 16 mm LAS16 / Adafruit-style clones) have **five Faston tabs**, not two. The five functions are always the same even when the silk is missing:

Tab	What it is
C	Switch common
NO	Normally open — closes to C while pressed

Tab	What it is
NC	Normally closed — opens from C while pressed (leave unused)
LED +	Ring anode (this listing is 5 V , built-in resistor on most 3–6 V rings)
LED –	Ring cathode

The switch and the LED are **separate circuits**. The kiosk only reads the switch (internal pull-up, press = GND). The rings are wired **always on** so the colours stay visible in a dim shop.

Identify the tabs with a meter (pin order on the back is not the same on every clone):

1. The two tabs that are a **diode**, not a click: that pair is **LED + / –**. One way around they drop ~1.8–3 V; the other way they are open. Tabs marked + / – (or a different colour / bent pair) are this pair.
2. The remaining three are the switch. **Unpressed: C–NC** beeps, **C–NO** is open. **Pressed: C–NO** beeps, **C–NC** is open. The tab that beeps in both tests (one at a time) is **C**.

Most likely rear view (looking at the **back**, tabs toward you) on these 16 mm 5 V rings:

+	–	← LED (often the two odd / marked tabs)	
NO	C	NC	← 2.8 mm Faston. NC unused.

If that drawing does not match the part in your hand, trust the meter, not the sketch.

Kiosk wiring (one button):

GPIO (BCM, pull-up)	— NO
GND	— C
NC	unused
5 V (header pin 2)	— LED +
GND	— LED –

Same pattern on all three. Share **5 V** and **GND**. Do **not** jumper LED + onto a GPIO, and do **not** put 5 V on C / NO / NC.

If	Then
Ring stays dark	Swap LED + and –
Ring is dim	Listing is 5 V — use header 5 V , not 3.3 V
Ring gets hot	Add ~220 Ω in series with LED + (that clone has no built-in resistor)
Press does nothing, GPIO error	5 V reached a GPIO — move the switch onto C/NO only, LED stays on 5 V

Panel layout (buttons **below** the screen, left→right as you face it). The colour chips on the bottom of the UI are in this order:

Left	Middle	Right
Green LOAD (short / operator)	Yellow SET (setter)	Red RUN (full)
GPIO 23 / header 16	GPIO 24 / header 18	GPIO 22 / header 15

2.4 Wiring diagram (defaults)

Pi 5 40-pin header (same BCM layout as Pi 3/4), looking at the board with the USB-A / Ethernet ports down:

3.3V	(1)	(2)	5V	← encoder VCC to pin 1; LED rings to pin 2
GPIO2	(3)	(4)	5V	
GPIO3	(5)	(6)	GND	← shared GND (encoders + buttons + LED -)
GPIO4	(7)	(8)	GPIO14	
GND	(9)	(10)	GPIO15	
GPIO17	(11)	(12)	GPIO18	← file encoder A (CLK)
GPIO27	(13)	(14)	GND	← file encoder B (DT)
GPIO22	(15)	(16)	GPIO23	← red RUN green LOAD
3.3V	(17)	(18)	GPIO24	← yellow SET
GPIO10	(19)	(20)	GND	
GPIO9	(21)	(22)	GPIO25	← optional sleep/wake (leave open if unused)
...				
GPIO5	(29)	(30)	GND	← mill encoder A (CLK)
GPIO6	(31)	(32)	GPIO12	← mill encoder B (DT)

DFRobot Fermion EC11 (SEN0235) silk:

EC11	Pi
VCC	pin 1 (3.3 V)
GND	pin 6 (GND)
A	pin 11 (GPIO 17) file, or pin 29 (GPIO 5) mill
B	pin 13 (GPIO 27) file, or pin 31 (GPIO 6) mill
C	not used

If turning the **file** knob moves the highlight the wrong way, use **Reverse** on the file knob in settings (or `encoder_swap = true`, or swap A and B). Same for the mill knob (`encoder_mill_swap`). SEN0235 is **20 pulses** per turn (one detent per pulse). **Ticks per tooth** is shared: GPIO ticks from one rest valley to the next. Start at **2** if one click skips two files. The list (or mill) changes halfway, so a small wiggle at rest does not skip. The file list and the mill list **wrap**: past the last name is the first, past the first is the last.

2.5 Screen orientation

The panel is 800×600 landscape electronics, mounted as portrait. The OS must present **600×800**. On Raspberry Pi OS the default is **Wayland**. Switch to **X11** (tkinter + screensaver `xset` are unreliable on Wayland/labwc):

```
sudo raspi-config
```

- **Advanced Options** → **Wayland** → **X11**
- **System Options** → **Boot / Auto Login** → **Desktop autologin**
- Reboot

Then rotate. Either:

Screen Configuration (desktop): HDMI output → Orientation **right** or **left** until the picture matches the physical panel. Apply and "OK".
Or in a terminal after login:

```
xrandr --output HDMI-A-1 --rotate right
```

(Use `xrandr` with no arguments to see the output name: HDMI-A-1, HDMI-1, Pi 5 KMS is usually HDMI-A-1 for HDMI0.)

To make rotation survive reboot, add the same `xrandr` line to `~/.config/autostart/` or `/etc/xdg/lxsession/LXDE-pi/autostart`.

Do not use legacy `display_rotate=` in `/boot/firmware/config.txt`. Pi 5 is KMS-only.

2.6 Printer (MUNBYN P047)

USB to the Pi. The kiosk sends **raw ESC/POS**: optional STEP bitmaps (gs v 0), then 48-column Font A, then cut. Do **not** print HTML or use a Windows GDI/POS-80 raster driver.

The app writes that blob to `/dev/usb/lp0` first (same as `open("/dev/usb/lp0", "wb").write(...)`). CUPS is not involved on a working Pi. Named CUPS (`lp -d munbyn -o raw`) is only a fallback if the USB node is missing or busy. There is no `lp` default-queue attempt.

If CUPS has already claimed the printer, the USB write fails with *Device or resource busy*. Stop or disable that queue (or CUPS) so `usb1p` owns `/dev/usb/lp0`.

Before RUN / LOAD / SET the kiosk sends **DLE EOT** (real-time status, not printed) on that same node. Cover open, paper end, or cutter error stay on the status line (**Pokrywa otwarta / Brak papieru / Zacięcie drukarki**) and the ticket is not sent. If the firmware does not answer, print goes ahead as before.

2.7 Optional sleep / wake button (GPIO)

Not required. The shop kiosk blanks after `idle_seconds` (default 60) and wakes on either encoder, a USB stick, or a keyboard / mouse whether this button is fitted or not.

If you add a fourth vandal, wire it like LOAD:

```
GPIO 25 (header 22, pull-up) — NO
GND (common with the other buttons) — C
NC unused
LED + / - — same 5 V / GND as the print rings, if the part has a ring
```

Press while the screen is on: DPMS off (panel black, Pi still running). Press again: wake. LOAD / SET / RUN still do not wake (avoids a ticket in the dark). An unconnected GPIO 25 stays high on the internal pull-up and never fires. If opening that pin fails, knobs and print buttons stay up.

Do **not** tie this GPIO to Pi 5 **J2**. Sleep is software DPMS; J2 is a real power switch.

2.8 Optional Pi 5 power button (J2)

Pi 5 already has a power button on the PCB. **J2** (two pads, silk **PWR_BTN**, between the RTC battery connector and the board edge) is the same switch in parallel. A panel button is a momentary **NO** across **those two pads only**.

Do	Do not
NO across the two J2 pads	Tie J2 to GPIO 25, BCM 20, or GPIO GND
Leave J2 open if you skip the panel switch	Expect Pi 4 WAKE_ON_GPIO / gpio-shutdown on GPIO3 — that is not how Pi 5 power works
Keep the PCB button reachable for service	Put 5 V LED rings on J2

Without J2 wired the Pi behaves as stock: plug in USB-C and it boots. Halt from the desktop or `sudo halt` still works. Unplug / replug USB-C to start again.

Short press (OS running) generates Linux `KEY_POWER`. Raspberry Pi Desktop then shows Shutdown / Reboot — wrong on a shop panel. Optional, only if you want the PCB or J2 button to halt with no dialog:

```
sudo nano /etc/systemd/logind.conf
```

```
HandlePowerKey=poweroff  
HandlePowerKeyLongPress=poweroff
```

Then `sudo systemctl restart systemd-logind` (or reboot). This affects the **onboard** button as well as J2. Omit the file and the kiosk still runs; you just get the Desktop dialog if someone presses the PCB button.

Short press after halt / PMIC standby boots again. **Hold** a few seconds is a hard cut.

Optional EEPROM (lowest power after halt; still optional):

```
sudo -E rpi-eeprom-config --edit
```

```
POWER_OFF_ON_HALT=1
```

Then a clean halt puts the PMIC in standby; J2 or the PCB button wakes it. USB-C unplug / replug still applies 5 V and boots **unless** you also set `WAIT_FOR_POWER_BUTTON=1`. **Do not set** `WAIT_FOR_POWER_BUTTON` **unless J2 is on the panel** (or the PCB button stays reachable). Without a power button that flag leaves the Pi sitting dead after a power cut.

fh6parse does not read J2. 5 V LED rings on the 40-pin header stay on after halt unless you switch that 5 V yourself.

2.9 GPIO screw terminal and wires

GPIO adapter (Pi end). GPIO screw terminal board — [Kamami 588019](#) (same 40-pin screw block as the 52Pi listing). Sold for Pi 4B / 3B+ / Zero; the Pi 5 **40-pin header is the same**, pin 1 still at the USB-C end. Press the female 2×20 onto the header. Fit the kit **M2.5 brass standoffs** so vibration cannot walk it off. Official Pi 5 cooler stays on the SoC; this board only occupies the GPIO strip. Seat the **FAN** cable first. If the terminal still will not sit (cooler, FAN plug, or case lid), put a **40-pin GPIO riser** (stacking header, extra-tall female-to-male) on the Pi first, then the screw terminal on the riser. Same pin 1 / BCM map — do not rotate the riser. The **NVMe Base Duo** is **under** the Pi (**\$2.10**), not stacked on this header. If the parcel is the larger LED HAT (52Pi EP-0129), it can fight that cooler — keep the cooler, do not stack a second HAT.

Silk is **BCM**: I017 = GPIO 17, not header pin 17. Confirm pin 1 before the first screw. Leave **GPIO 2 / 3** (header 3 / 5, I²C1) empty for a future Qwiic keypad. **J2 is not on this board**. Colour card: [LINUX-KIOSK-WIRING.pdf](#) page 4.

Three looms, not one bag of jumper wires. GPIO sense, LED rings, and Pi power must never share a conductor that can put 5 V on a BCM pin. **Do not** use loose Dupont jumper packs as the finished harness.

Buy

Circuit	Wire	Ends
Encoder A/B, button NO (3.3 V sense)	24 AWG stranded (0.25 mm ²), PVC or silicone	Screw on the 52Pi board; Dupont or solder at the EC11
Encoder VCC 3.3 V , common GND	22 AWG stranded (0.34 mm ²)	Screw on 3.3 V (pin 1 / 17) and GND
LED + 5 V and LED –	22 AWG stranded	Screw on 5 V (pin 2) / GND; insulated 2.8 mm Faston on the vandal
Optional J2 power	26–28 AWG , keep under 20 cm	Solder or JST-SH 1.0 2-pin on J2 only
Future Qwiic keypad	Official Qwiic 4-pin cable	Do not steal SDA/SCL screws for buttons

24 AWG is the default for every GPIO run under ~40 cm. **22 AWG** for 3.3 V / 5 V / GND so a 2.8 mm Faston crimp holds. Terminals take about **16–26 AWG**; stay 22/24 in the loom. Stranded only (vibration). Bootlace ferrule on the screw end; do not tin a blob. Silicone is nicer next to the Pi 5 cooler; PVC is fine in the rest of a 3D-printed case.

Colour (so a meter is not required at 2 a.m.)

Colour	Net
Black	GND (one net: knobs, C tabs, LED –)
Orange	3.3 V encoder VCC (pin 1 / 17)
Red	5 V LED + only (pin 2)
Blue / light blue	File encoder A / B
White / grey	Mill encoder A / B
Green	LOAD NO (BCM 23)
Yellow	SET NO (BCM 24)
Brown	RUN NO (BCM 22) — not red
Violet	SLEEP NO (BCM 25), if fitted
Grey twisted pair	J2 only

Do not use the same red for RUN and for LED +. That mix-up is a dead Pi.

Terminations. Pi: screw terminal on the 40-pin header, or on a **GPIO riser** if it will not fit next to the cooler (standoffs, strip ~5 mm, ferrule). Strain-relieve the bundle to the case, not to the board. Vandal: insulated 2.8 mm Faston females, **crimped** (do not solder on the switch). Leave **NC** empty. EC11: Dupont or solder on the pin row; leave shaft **C** open. J2: two short flying leads across the pads only — never a screw on the GPIO block.

Length. GPIO A/B and button NO **under ~40 cm**. Twist each encoder pair. Do not tape encoder wires along the 5 V LED bundle or a USB 3 cable. Star GND at header pins 6 / 9 / 14 — do not daisy LED – through a switch C. LED rings are tens of mA; pin 2 is enough. The official **USB-C 5 V / 5 A** cable is the only high-current wire. Do not jumper the NVMe Duo extra 5 V pads onto pin 2 unless a drive actually browns out (**\$2.10**).

2.10 NVMe Base Duo (PCIe, not GPIO)

Pimoroni **NVMe Base Duo** (PIM704) — [Botland](#). Two M.2 **M-key** NVMe slots (2230–2280) on the Pi 5 **PCIe FPC** (next to the PCB power button). It is not a 40-pin HAT. The 52Pi screw terminal, official cooler, and J2 stay on **top** of the Pi.

Mount under the Pi. Kit 12 mm M2.5 standoffs through the four Pi holes. Power **off** first. FPC: wider **ADDON** end into the Duo (grey clip flips up; writing faces down into the socket), **RPI 5** end into the Pi (brown clip slides ~1 mm). Pirate logo / writing faces out when folded. Fold the Duo under the Pi like a hinge — do not crease the flex. Assembly: [Pimoroni getting started](#).

The flex can cover the **microSD** slot. Insert a rescue SD **before** folding. Shop kiosk: boot from **slot A** NVMe once firmware is **2024-05-17 or newer**:

```
sudo apt update && sudo apt upgrade
sudo reboot
sudo rpi-eeeprom-update
lsblk
```


lsblk must show nvme0n1. Then Raspberry Pi Imager onto that disk, and raspi-config → Advanced Options → Boot Order → **NVMe/USB Boot**. Git clone in **\$3.2** then lives on that disk. USB stick still carries .nc / .stp.

Disk: official Raspberry Pi **512 GB NVMe, 2230** (22 × 30 mm), M-key, TLC, 3.3 V, max **2.8 W**. Buy the [Raspberry Pi SSD Kit 512 GB](#) and **take only the SSD**. The kit HAT+, 16 mm GPIO spacers, and screws stay in the drawer — that HAT+ would sit on the 40-pin header and fight the 52Pi terminal and cooler. On the Duo, the SSD is already on the official HAT+: undo the tiny M2, plug it into **slot A**, and use the Duo's **2230** standoff hole (closest to the socket), not 2280.

One drive. Leave slot B empty (not RAID, not required for fh6parse). Stay **PCIe Gen 2** even though the Pi SSD is rated Gen3×4; do not set dtparam=pciex1_gen=3. Official **27 W** PSU. Extra 5 V pads on the Duo stay unused at 2.8 W.

2.11 Panel USB 3.0 (pendrive)

Metal **USB 3.0 Type-A** socket, **27 mm** panel cutout — [Allegro](#). Screw it onto the 3D-printed enclosure (nut on the inside). The operator plugs the stick into the front; a short USB 3 pigtail runs to a Pi **USB 3** port (the blue pair next to Ethernet). Not GPIO. Not power for the Pi or the P047.

Keep the P047 on a **separate** Pi USB-A (USB 2 is enough). Do not daisy the printer through this panel socket. Strain-relieve the pigtail to the case, not to the Pi header. Do not tape it along encoder A/B. The kiosk still automounts under /media / /run/media — no extra driver.

2.12 Active Cooler

Official Raspberry Pi **Active Cooler** (heatsink + PWM fan) — [Botland RPI-23925](#). Pi 5 only. Clip the heatsink onto the SoC; plug the **4-pin** lead into the **FAN** header next to GPIO pin 1 (VCC, GND, PWM, tach). Firmware drives the fan; fh6parse does not. Compatible with the NVMe Duo **under** the Pi. Seat this cable **before** the GPIO screw terminal. If the terminal still will not clear the cooler, use a GPIO riser (**\$2.9**). The spring clips are not meant for repeated removal. Print the case in **PETG** with vents above this fan and below the Duo (**\$2.13**).

2.13 Must do and shop suggestions

Same tables: [HARDWARE.md](#). The parts list is coherent; shop failures are **connectors and heat**, not missing SKUs.

Must do at assemble

Do	If you skip it
Never put 5 V on a BCM pin. LED rings on pin 2 only. RUN is brown , not red.	Dead Pi
Seat the PCIe flex with power off, both clips locked, no crease. The four Duo standoffs carry the sandwich. Keep a spare PIM702 / PIM703 in the drawer.	Black screen; OS is on that cable; SD slot often blocked
Clamp USB-C and micro-HDMI0 to the case. Official 27 W cable only — no USB-C extension (PD 5 A dies).	Undervoltage / blank panel after vibration
Screw-terminal silk is BCM (I017 = GPIO 17, not header pin 17). Pin 1 at USB-C. Do not rotate a riser.	3.3 V on a GPIO or one-pin-over short
PETG case, vents above the Active Cooler and below the Duo. Do not pack foam on the blower.	Thermal throttle / PLA warp
Strain-relieve the GPIO bundle and the USB 3 pigtail to the case , not the Pi. Thick boss + nut/washer on the 27 mm USB hole.	Walked header / cracked print

Suggestions (first month in the shop)

- Keep the USB 3 pigtail off the encoder loom (USB 3 is noisy).
- Solder or glue the EC11 Dupont end; the Pi end is already screws.
- Cable-tie a GPIO riser so it cannot lift.
- Threadlocker on the 2230 M2; check after a week of vibration.
- Dummy USB plug when idle; coarse foam on the cooler intake (CNC dust).
- 27 W brick close to the Pi.

Leave unless it bites: J2 halt, UPS, slot B clone, PCIe Gen 3, RAID. Git **UPDATE** does not rewrite the boot partition; a power cut during apt / EEPROM still can.

3. Software

Do this on the Pi, as user **kiosk**, with network.

3.1 Packages

```
sudo apt update
sudo apt install -y git python3 python3-pip python3-tk \
    python3-gpiozero python3-igpio python3-rpi-igpio \
    x11-xserver-utils
```

Do **not** install `python3-rpi.gpio` on a Pi 5. That library talks to the old SoC GPIO; Pi 5 GPIO is on RP1. `python3-igpio` is the driver; `python3-rpi-igpio` is only a compatibility shim. The kiosk prefers `igpio` automatically.

`python3-tk` is the kiosk UI. **Do not install CUPS** for the shop P047 — the kiosk prints by writing `/dev/usb/lp0`. `x11-xserver-utils` provides `xset` so the panel can blank.

Confirm Python is 3.10 or newer:

```
python3 --version
```

3.2 Install fh6parse

Recommended — from source (git). This is the shop update path.

```
cd /home/kiosk
git clone https://github.com/Don-Pablo-G/fh6parse.git
cd fh6parse
sudo pip3 install -e . --break-system-packages
```

For isometric views on the ticket, also install the CAD extra (**\$3.7**). On 64-bit Pi 5 this usually has wheels. If `cascadio` fails, skip it; tickets stay text-only.

```
sudo pip3 install -e '.[models]' --break-system-packages
```

`--break-system-packages` is normal on Bookworm when you are not using a venv. `gpiozero` stays the **apt** copy so it can see the Pi GPIO.

Allow user **kiosk** to restart the kiosk after an on-screen **UPDATE** (once, as root):

```
echo 'kiosk ALL=(root) NOPASSWD: /usr/bin/systemctl restart fh6parse-kiosk' | sudo tee /etc/sudoers.d/fh6parse-kiosk
sudo chmod 440 /etc/sudoers.d/fh6parse-kiosk
```

Check:

```
python3 -m fh6parse --version
```

Expect fh6parse 1.4.0+..... (package number plus git short SHA). If the number is older, this clone is behind — `git fetch && git pull --ff-only` then check again (**\$8**).

Later upgrades are **\$8**. Do not run `pip install` on every pull. The kiosk does not update by itself.

Air-gap first copy — prebuilt one-file (this repo's `dist/packages`). Not the upgrade path; use git for later updates.

On a Pi 5 use the **aarch64** tarball only (there is no 32-bit Raspberry Pi OS for Pi 5).

```
cd /home/kiosk
tar -xzf fh6parse-*-raspberrypi-aarch64.tar.gz
chmod +x fh6parse
sudo cp fh6parse-kiosk.ini.example /etc/fh6parse-kiosk.ini
```

Run:

```
./fh6parse --kiosk --config /etc/fh6parse-kiosk.ini
```

The one-file ARM tarball does not bundle the CAD stack, so **\$3.7** pictures are git-checkout only. There is no **UPDATE** button and `--update` refuses this install. To get the 1.4.0 shop update path later, switch to the git checkout above.

For systemd, set `ExecStart=/home/kiosk/fh6parse --kiosk --config /etc/fh6parse-kiosk.ini` (path to the unpacked binary).

3.3 Kiosk config

```
sudo cp /home/kiosk/fh6parse/packaging/fh6parse-kiosk.ini.example /etc/fh6parse-kiosk.ini
sudo nano /etc/fh6parse-kiosk.ini
```

Leave the defaults unless your wiring or printer queue differs. Useful keys:

Key	Default	Meaning
idle_seconds	60	Black screen after this many seconds. 0 disables.
encoder_clk / encoder_dt	17 / 27	File knob BCM pins (EC11 A / B). Change in settings or here.
encoder_mill_clk / encoder_mill_dt	5 / 6	Mill knob BCM pins (header 29 / 31).
encoder_swap / encoder_mill_swap	false	Reverse file knob / mill knob if the list moves the wrong way. Same as Reverse in settings.
encoder_steps	1	GPIO ticks from one tooth valley to the next (shared). DFRobot EC11 is usually 2 . Highlight / mill changes at half a tooth so rest is stable.
button_run / button_load / button_set	22 / 23 / 24	Red RUN / green LOAD / yellow SET. Old <code>button_full</code> / <code>button_min</code> still read.

Key	Default	Meaning
button_spare	25	Optional sleep/wake. Press blanks the panel; press again wakes. Unwired pin never fires.
button_delay	2	Seconds to ignore LOAD / SET / RUN after a ticket so a double press does not print two slips. 0 = no wait. Change in settings .
printer_device	/dev/usb/lp0	USB printer node (tried first)
printer_queue	(empty)	Optional CUPS name; used only if the USB node fails. Empty = never call lp.
scan_depth	1	USB root only. Raise to search subfolders.
extensions	.nc, .tap	File types (case-insensitive)
extra_roots	(empty)	Extra folders to list, comma-separated (for testing)
model_roots	(empty)	Optional company .stp folders. USB stick is searched first. See §3.7 .
language	p1	Screen language: p1 (default) or en. Change on the kiosk in settings (F2 / C , or click PL / EN). GPIO pins, mill, and encoder ticks are on the same panel. Stored in ~/ .config/fh6parse/ui. ini (and in this ini if it is writable).
machine	default	Id of the mill used for cycle time ([machine.<id>] below). Change in settings .
fullscreen	true	Shop display. Escape once exits fullscreen.

Add one [machine.<id>] section per mill, or use **Add mill...** on the Windows/Linux GUI and in kiosk **settings**. Built-in **Default mill** is 20 m/min rapids and 0 s tool change until you pick another. Keys:

Key	Default	Meaning
name	the id	Label on the kiosk, Windows GUI, and tickets
rapid_mm_min	20000	Linear G0 rate (mm/min) for XYZ time
rotary_deg_min	5400	B/C G0 rate (deg/min)
tool_change_s	0	Seconds added at every Txx M6 (carousel swap only)
max_rpm	(omit)	Spindle limit. A tool's S above this prints a WARNING on LOAD / RUN.
atc_x atc_y atc_z	(omit)	That mill's tool-change position in G53 mm (one field per axis). B and C are 0 . Old atc_b / atc_c still read.
offset_x offset_y offset_z	(omit)	Typical vise/table work origin in G53 mm (G54, G55, ... — the same machine point). B and C are 0 . Old g54_x / g54_y / g54_z still read.
tool_length_mm	0	Approximate stick-out (Z only). Haas H is a register, not mm.
x_min x_max y_min y_max z_min z_max	(omit)	Machine travel envelope in G53 mm, one field per axis and limit. Tickets then show where the work offset may sit so programmed XY stays inside travel.

If ATC and offset XYZ are all set, cycle time uses one G53 pose: work rapids convert through the stored offset + tool length; at Txx M6 the finishing tool rapids Z then XY to that mill's ATC (B0 C0), then tool_change_s on the new T. A program that already g53's to the ATC is not charged twice. Omit the keys to keep the older estimate (Default mill).

If XY travel is set, the ticket adds a labeled rectangle of allowed work-offset origin (G53 mm): four corners, center, max Ø for centred outside G41/G42 (the shorter leftover), and a warning if the stored offset is outside or the work is larger than travel. Needs Pillow for the PNG; numbers still print without it.

Example:

```
machine = vf-4ss

[machine.vf-4ss]
name = Haas VF-4SS
```

```

rapid_mm_min = 25400
rotary_deg_min = 5400
tool_change_s = 2.8
atc_x = -750
atc_y = -20
atc_z = 0
offset_x = -400
offset_y = -250
offset_z = -400
tool_length_mm = 120
x_min = -1270
x_max = 0
y_min = -508
y_max = 0
z_min = -635
z_max = 0

```

3.4 Groups and devices

```
sudo usermod -aG gpio,lp kiosk
```

Log out and back in (or reboot) so the groups apply. Printer node check is **\$3.5**.

3.5 Direct USB printer (recommended)

No CUPS. After the P047 is plugged in:

```

lsusb
ls -l /dev/usb/lp0

```

User kiosk must be able to write the node (lp group, then log out / reboot). Test the same write the kiosk uses:

```

python3 - <<'PY'
from pathlib import Path
Path("/dev/usb/lp0").write_bytes(b"\x1b@TEST\n\n\n\x1dVB\x00")
PY

```

You should get a short slip and a cut. If you see *Permission denied*, the group has not applied yet. If you see *Device or resource busy*, something else (usually CUPS) has the printer — `sudo systemctl stop cups` and try again, then disable CUPS so it does not come back:

```
sudo systemctl disable --now cups
```

If `/dev/usb/lp0` is missing, unplug/replug and look at `dmesg | tail`. A second printer can appear as `/dev/usb/lp1` — set `printer_device` in the ini.

CUPS is not required. Leave `printer_queue` empty to never call lp. Only set a raw queue if the USB node never appears on that machine.

3.6 First run (with keyboard)

On the graphical desktop:

```
python3 -m fh6parse --kiosk --config /etc/fh6parse-kiosk.ini
```

Without GPIO you can still use a **USB keyboard and mouse** at any time (hot-plug is fine). The kiosk keeps keyboard focus and the black screensaver wakes on a key, click, or mouse wheel.

The shop screen is **Polish** unless `language = en` is set. Open **settings** with the keyboard or mouse (not the encoder): **F2** or **C**, or click the **PL / EN** chip next to the version. Pick **Polski** or **English**, the mill (rapids, tool-change time, optional G53 ATC / work offset / tool length / travel from `[machine.<id>]` in this ini; the mill name also sits next to the version chip and the mill knob cycles it), **Add mill...** to create a new mill (written to `~/.config/fh6parse/ui.ini`), a **LOAD | SET | RUN** tick matrix for ticket content (Reset LOAD / SET / RUN restore the factory packs; G68 / D vs T / S max / late offset / G95 / travel-too-big always print), BCM pin numbers for file A / B, mill A / B, green LOAD / yellow SET / red RUN / optional sleep, knob **Reverse**, **ticks per tooth**, and **print wait** after a ticket (GPIO ticks from one rest valley to the next; the highlight or mill changes halfway so a wiggle at rest does not skip; file and mill lists wrap). Language, mill, GPIO, and `report_load / report_set / report_run` are written to `~/.config/fh6parse/ui.ini` (user kiosk can write this even when `/etc/fh6parse-kiosk.ini` is root-owned) and, if permitted, into the main ini. Pin changes take effect immediately (GPIO is reopened). **Esc** closes the mill form first, then settings; the next **Esc** still leaves fullscreen. The file encoder or a GPIO print button closes settings without printing / skipping a file. The mill encoder keeps settings open and changes the mill.

Input	While awake	While screensaver
Arrows, mouse wheel, click a file	Move highlight	First event only wakes
F / M / S	Print RUN / LOAD / SET	Ignored (no ticket); another key or click wakes
GPIO RUN / LOAD / SET (red / green / yellow)	Print	Ignored (no ticket, stays black)
GPIO spare (optional)	Sleep (black panel)	Wake
Yellow UPDATE to ... / U	One tap: pull, then restart kiosk	Wake first, then tap
F2 / C / click PL-EN	Open or close settings (language, report ticks, pins, encoder ticks)	Wake first
In settings: arrows / wheel / Polski-English	Switch language	—
In settings: + / -	BCM pins, ticks per tooth, and print wait	—
Esc	Close settings, else leave fullscreen, then close	Wake, then Esc again leaves fullscreen

Plug in a USB stick with `.nc` files; the list should fill by itself.

Desktop autostart of the kiosk is in the next section. Until then, Escape leaves fullscreen, Escape again closes the window.

3.7 STEP models on the ticket

Optional. The kiosk looks for a matching `.stp` / `.step` **on the USB stick first** (same folder as the `.nc`, then subfolders on that stick). No NAS is required. Company folders in `model_roots` are a fallback if the stick has no match.

SET and RUN tickets can show two opposite **true isometric** views (45° then ~35.3° — look along the cube diagonal), as **visible edges only** (silhouette + sharp creases, no shading). Through-holes draw as ellipses (near rim, and the far rim only where you can see through). The longest 3D axis is laid across the 80 mm width (~512 dots); height is cropped to the part, so a long thin shaft is a thin strip, not a metre of paper. A bulky part is capped (~30 mm of paper per view). Print never waits for a model. After an update, delete `/tmp/fh6parse-models` so old shaded bitmaps are not reused.

1. USB (usual shop path). Put the STEP file next to the program, or in a subfolder on the same stick:

```
D0134078.nc
D0134078_Rev03.stp
```

or D0134078.nc plus cad/D0134078_Rev03.stp. The kiosk indexes the stick when it is inserted. Matching rules are **§3.7 step 4**. CAD extra (**step 3**) is still required to draw the picture.

2. Optional company CAD folders in /etc/fh6parse-kiosk.ini if the stick has no STEP. Several roots are allowed (comma or : / ;). Subfolders are searched. A file on the stick always wins over the NAS.

```
model_roots = /mnt/fh6parse-cad
```

fh6parse **only reads** that folder (list .stp / .step, open for the isometric). It never creates, overwrites, or deletes files there. Rendered bitmaps go to /tmp/fh6parse-models on the Pi (or %TEMP%\fh6parse-models on Windows). Reports are not written next to company files. CIFS is mounted **read-only**, so the kernel also refuses writes. Network shares (CIFS/NFS) are never treated as a USB stick.

Easy mount of the office Z: drive (plug the Pi into company Ethernet). On a Windows PC, net use Z: shows the UNC (example \\fileserver\Dokumentacja). Then on the Pi:

```
sudo apt install -y cifs-utils
sudo bash /home/kiosk/fh6parse/packaging/connect-windows-share.sh //fileserver/Dokumentacja --guest
```

Domain account (more common):

```
sudo bash /home/kiosk/fh6parse/packaging/connect-windows-share.sh //fileserver/Dokumentacja --user kiosk
sudo nano /etc/fh6parse-cad.cred
```

Put the password on the password= line, chmod 600, then sudo mount /mnt/fh6parse-cad. The script writes a **read-only** fstab line with x-systemd.automount: when the cable is unplugged the kiosk still starts; when you plug in, the share appears on first use (STEP cubes fill in within a minute). Template: packaging/fh6parse-cad.cred.example.

Manual equivalent:

```
sudo mkdir -p /mnt/fh6parse-cad
# /etc/fstab (one line):
# //fileserver/Dokumentacja /mnt/fh6parse-cad cifs credentials=/etc/fh6parse-cad.cred,ro,uid=kiosk,gid=k:
```

If the share is down, the ticket is still text only.

On **Windows**, pick **STEP folders...** once. Prefer the UNC (\\fileserver\Dokumentacja) so it still works if the letter is not Z:. That path is model_roots in fh6parse-kiosk.ini. The GUI will not save reports into that folder.

3. Install the CAD extra (git checkout only; heavy: numpy, pillow, trimesh, cascadio):

```
sudo pip3 install -e '[models]' --break-system-packages
```

If that install fails, leave it off. Matching still runs; nothing is rendered and the list icon stays off.

4. Matching starts at the first characters of the NC file name (and the o program title). The end of the STEP name may differ (_Rev03, _0P1, extra words). Nearby part numbers do not match (D0134078 will not pick D0134079).

Revision is taken from the G-code header, then the title, then the file name:

In the NC	What is used
(REV 3), (REV.03), (REVISION A)	that revision
(Rewizja: 02), (REW 2), (WERSJA 1)	that revision
001282 (SE0241282-0 WIERCENIA ...)	part SE0241282, rev 0
D0134078_Rev03.nc / D0134078-R2.nc / D0134078A.nc	suffix on the name
No rev in header or name	latest matching .stp / .step

If the program **has** a revision, only a STEP file with the **same** rev is used (not a newer one). If several files share that rev, the closest (shortest) name wins.

NC	Header	Picked model
D0134078.nc	004078 (D0134078) (no REV)	D0134078_Rev03.stp over _Rev02
D0134078.nc	(REV 2)	D0134078_Rev02.stp, not Rev03
SE0241282.nc	001282 (SE0241282-0 ...)	SE0241282-0.stp (or _Rev0)
000814086.nc	title 000814086 OP1/OP2	000814086_Rev02.stp if that is latest

5. On the screen, a small **wireframe 3D cube** appears next to the file when the bitmap is rendered and ready (same icon as the legend under the title — the same visible-edge isometric language as the ticket). The **stacked isometric** itself is shown under the highlight preview (and above the Windows GUI report) so a wrong STEP can be caught before print. The walk and render run in the background for every USB file in the list. Cache: /tmp/fh6parse-models.

On **Windows**, the GUI also searches next to the opened NC file. **STEP folders...** is the company-share fallback (prefer \\server\share, not only Z:). Paths are saved as model_roots in fh6parse-kiosk.ini next to the exe. fh6parse will not write reports into that folder. The Windows one-file build bundles the CAD stack; print still works if a model is missing.

4. Boot to kiosk

Copy the unit and enable it. The service assumes user kiosk, display :0, and Desktop autologin. If model_roots is on a company share, use the read-only automount in **§3.7** (/mnt/fh6parse-cad). The kiosk starts even if the cable is unplugged; cubes appear after the share is up. The sudoers line from **§3.2** must exist if you want the **UPDATE** button to restart the unit.

```
sudo cp /home/kiosk/fh6parse/packaging/fh6parse-kiosk.service /etc/systemd/system/  
sudo systemctl daemon-reload  
sudo systemctl enable fh6parse-kiosk.service
```

Optional: turn off the *desktop* screensaver so it does not fight the app. In /etc/xdg/lxsession/LXDE-pi/autostart (path may vary) add:

```
@xset s off  
@xset -dpms
```

The kiosk still blanks after 60 s via a black overlay and xset dpms force off.

Reboot. After the desktop appears, the CNC list should open fullscreen.

Logs:

If the unit starts before X is ready, it will restart every 3 s until :0 exists.

5. Daily use

1. Power on. Screen shows **Włóż pendrive / Insert USB** (or the last stick if it was already plugged in). The kiosk is Polish unless settings were changed.
2. Insert the USB stick. `.nc` / `.tap` files in the stick **root** appear. The count line shows **Można wyjąć / Safe to remove** when the kiosk is not reading the stick — wait for that before unplugging. **Czytanie pendrive — czekaj / Reading USB — wait** means preview parse or a STEP copy from the stick is still running.
3. Turn the **file** encoder to highlight a file (past the last name wraps to the first). The panel under the list shows each operation's tool count and cycle time, and whether the STEP views are ready. A **3D cube** next to the name also means the STEP views are ready; the stacked isometric appears under the preview when it is. Check OP1 vs OP2 here before printing. If that `.nc` is overwritten on the stick (same name, new bytes), preview reloads from disk — wait for **Reading...** to finish before **RUN / LOAD / SET**. The mill name sits next to the version chip; turn the **mill** knob to change mill (same wrap as **Add mill...** / settings).
4. **Green LOAD** (left under the screen) — factory operator slip: file / program / units, tool list with T / H / D / S / Min Z and load boxes, always-on safety, sign-off. No mill, no cycle chart, no each-Txx-M6 dump, no offset box, no STEP. Change ticks in **F2**.
5. **Yellow SET** (middle) — factory setter slip: file / program, mill name, STEP when ready, offset rectangle + Ømax + Z window, cycle time (no share chart), programmer ! notes, always-on safety, sign-off.
6. **Red RUN** (right) — factory full slip: STEP, offset block, ops, cycle + time split (G0 / cut / rot / canned / probe / ATC stacked per T), tool list, each Txx M6 / M00, extra warnings. No load boxes, no sign-off. The three colour chips on the bottom of the screen sit in this same left→right order. After a ticket, **LOAD / SET / RUN** are ignored for **print wait** (default 2 s; **Czekaj przed następnym biletem / Wait before the next ticket**) so a second press does not cut two slips.
7. If the status line says **Pokrywa otwarta / Printer cover open, Brak papieru / No paper**, or **Zacięcie drukarki / Printer jam**, fix the P047 first. **RUN / LOAD / SET** will not cut a blank slip. Those messages also appear on their own while idle (polled about every 2 s).
8. If there is no cube, print anyway. The slip is text only. The chip next to the cube legend says why: **3D ready, searching...** / **szuka...** (looking for a `.stp`), **rendering...** / **liczy...** (drawing a match), **no STEP / brak STEP, Z: off / Z: wyłą.** (company share down), or **no CAD / brak CAD** (install [models]).
9. Status after a good print: `device:/dev/usb/lp0`. If it says `lp:...`, CUPS took the job — **\$3.5**.
10. **WARNING:** lines: `H{n}` does not match `T{tool}` on G43, `D{n}` does not match `T{tool}` on any D, G95 still active... if feed-per-rev was not cancelled with G94 before the next tool (or M30), empty-pocket (Txx M6 with no motion) except the **last** tool change (spindle prep), G55 after operation started (L...) if G54–G59 appears after the first Txx M6 or M97/M98 (offset belongs in the file preamble only), `S{n}` exceeds mill max {rpm} when that mill's max_rpm is set, and G68 T12 L40 to G69 T15 L80 / G68 T12 L40 without G69 when coordinate rotation is used (cancel with G69 before M30). Matching H/D stay quiet. Comments with ! print as **Programmer notes**.

11. After **60 seconds** with no encoder movement and no new USB, the screen goes black.
12. Wake: either encoder, inserting a USB stick, a **keyboard / mouse**, or the optional spare GPIO. The first encoder step, key, or click only wakes; it does not skip a file or print. GPIO print buttons while asleep stay ignored. Spare while awake blanks the panel (same as the 60 s idle). No spare button: idle timeout and knobs still work.
13. Settings: **F2 / C** or the **PL/EN** chip (mouse) — **§3.6**. The file encoder does not open settings. Mill picker, **Add mill...**, pins, knob reverse, ticks per tooth, and print wait are on that panel.
14. Print buttons **do nothing** while the screen is asleep (avoids accidental tickets).
15. Current version is **v1.4.0+.....** at the top right (mill name is next to it). The suffix is the git short SHA of this checkout. If the Pi is on the network and origin is ahead, a yellow **UPDATE to x.y.z** (or a git hash if the number is still 1.4.0) appears **after this boot's check**. It does **not** update by itself. One tap installs and **restarts** the kiosk (sudoers in **§3.2**). Print still works until you tap it.

Preview parse runs when a file is highlighted (idle, not on RUN / LOAD / SET). STEP matching and rendering run in the background and must not delay the ticket.

6. Troubleshooting

Symptom	What to check
GPIO off: ... on the status line	python3-gpiozero and python3-igpio installed (not python3-rpi.gpio on Pi 5); user in group gpio; pins not already claimed.
Knob does nothing	File knob defaults 17/27, mill knob 5/6. A/B (CLK/DT) on the BCM numbers shown in settings; common GND; 3.3 V VCC. Try Reverse for that knob.
Mill name does not change	Mill knob BCM 5/6 (header 29/31). Need more than one mill in settings. encoder_mill_swap if it turns the wrong way.
Knob skips or jitters at rest	Raise Ticks per tooth so the valley is several GPIO ticks wide; highlight only changes halfway to the next tooth. Shorter wires; module decoupling. The kiosk also sets a short encoder bounce_time for Pi 5. Loose jumper packs walk off the header — land wires on the screw terminal (§2.9) and fit the brass standoffs. Terminal will not seat next to the cooler: 40-pin GPIO riser, then the terminal on top.
Buttons print on press and release	Use momentary NO to GPIO and C to GND. Leave NC and the LED tabs off the GPIO. Not a latching switch. Raise print wait (button_delay) if two tickets still come out.
Czekaj przed następnym biletem / Wait before the next ticket	Normal after a print. Default 2 s (button_delay). Settings + / -.
Ring LED dark	LED + to header 5 V (pin 2), LED - to GND. Swap +/- if it stays dark. Dim on 3.3 V is expected on the 5 V rings.
Pi dies / GPIO error after wiring LEDs	5 V reached a GPIO . LED stays on pin 2; C / NO / NC must never see 5 V.
List stays on Insert USB / Włóż pendrive	Stick seated in the panel socket? Pigtail in a Pi USB 3 port? ls /media / ls /run/media. Format FAT32. Files ending .nc or .tap in the stick root .
Preview does not match the stick file	Same name overwritten? Wait until Reading... clears. Reload uses mtime and size (FAT 2 s). Do not yank during Reading USB — wait .
Yanked stick, list frozen / cube stuck	Wait for Safe to remove next time. Plug back in. rm -rf /tmp/fh6parse-models if pictures are from a half-copy.

Symptom	What to check
printer failed	<code>ls -l /dev/usb/lp0</code> ; user kiosk in group lp; test the Python write in \$3.5 . <i>Permission denied</i> → log out after <code>usermod</code> . <i>Busy</i> → CUPS still owns the printer (<code>sudo systemctl disable --now cups</code>).
Brak papieru / Pokrywa otwarta / Zacięcie drukarki	Load paper, close the cover, or clear the cutter. RUN / LOAD / SET is blocked on purpose. If the P047 never answers DLE EOT, print still goes through (no false alarm).
Blank slip / cut with no text	Paper was likely already out or the cover was open <i>before</i> this build. Confirm the status line.
Garbage on the slip	CUPS grabbed the job. The kiosk writes <code>/dev/usb/lp0</code> first; disable CUPS (\$3.5). Status must show <code>device:/dev/usb/lp0</code> , not <code>lp:....</code>
Ticket does not cut	Cutter empty/jammed. Status should show Zacięcie drukarki / Printer jam . App already sends ESC/POS cut (<code>GS V</code>) when status is OK.
Screen never sleeps	<code>idle_seconds = 0</code> , or encoder bouncing. Still on Wayland? Switch to X11 so <code>xset</code> works. Optional spare GPIO also blanks even when idle is 0.
Optional sleep button does nothing	Unwired is normal. If fitted: NO to GPIO 25, C to common GND, same as LOAD. Screen still blanks after 60 s without it.
Keyboard/mouse do nothing	Plug into the Pi USB-A; X11 picks them up. Click or press a key — the kiosk claims focus. F2 / C opens settings. Esc closes settings, then leaves fullscreen. GPIO print buttons still do not wake the screensaver.
Language resets to Polish after you picked English	Stored in <code>/home/kiosk/.config/fh6parse/ui.ini</code> . Pick English again in settings. UPDATE does not delete that file. Pins, mill, and ticks live in the same overlay.
Add mill... missing / mill list is only Default	Clone is older than this pull. Settings → Add mill... Saved in <code>ui.ini</code> (<code>[machine.<id>]</code>). Travel keys <code>x_min...z_max</code> print the offset rectangle.
Screen stays in English and F2 does nothing	Wake first if the screen is black. Click the EN chip next to v.... Encoder never opens settings.
Black screen immediately	Desktop blanking plus app DPMS. Disable LXDE idle blank; keep kiosk <code>idle_seconds = 60</code> .
Wrong aspect / sideways UI	Rotate until <code>xrpyinfo</code> (or Screen Configuration) shows 600×800. App geometry is 600×800 fullscreen. Pi 5 output is often HDMI-A-1.
Service dead, UI never starts	<code>echo \$DISPLAY</code> in a desktop terminal should be <code>:0</code> . <code>raspi-config</code> → X11, desktop autologin. <code>journalctl -u fh6parse-kiosk</code> . Unit User= must be kiosk.
Pi stays on after halt / no panel power	J2 is optional. USB-C unplug/replug boots. Do not set <code>WAIT_FOR_POWER_BUTTON=1</code> unless J2 is wired. PCB power button is the same as J2. See \$2.8 .
Undervoltage / random reboots	Official 27 W PSU, no USB-C extension , cable clamped. A phone charger or Pi 3 supply is not enough. The Pi 5 128 GB SSD is 2.8 W max; LED rings stay on the same PSU.
<code>lsblk</code> has no <code>nvme0n1</code>	Power off. Reseat both FPC clips: ADDON on the Duo, RPI 5 on the Pi. No crease in the flex. Firmware <code>sudo rpi-eeprom-update</code> dated 2024-05-17 or later. Official Pi 2230 SSD in Duo slot A (2230 hole, not 2280). Do not leave it on the kit M.2 HAT+. Spare flex: PIM702 / PIM703.
Blank / flickering LCD	micro-HDMI0 walked off — clamp it. Port next to USB-C.
Cannot reach the microSD slot	Normal with the Duo under the Pi. Boot from NVMe. Rescue SD goes in before folding the flex.
Python 3.9	Wrong image. Flash 64-bit Raspberry Pi OS Desktop for Pi 5.
<code>--update</code> says one-file package	This Pi is running the ARM tarball. Copy a new tarball or reinstall from git (\$3.2).

Symptom	What to check
--update / fast-forward failed	Uncommitted edits or a diverged branch. See §8.1 . Do not merge on the shop floor.
--update / UPDATE pulled but UI unchanged	sudo systemctl restart fh6parse-kiosk. Missing sudoers: §3.2 .
UPDATE button never appears	Offline, one-file tarball, already up to date. Check runs at start and again after the screensaver wakes. Screen should show v1.4.0 .
UPDATE says failed / kiosk did not restart	sudo -n systemctl restart fh6parse-kiosk from user kiosk should succeed after §3.2 . Then sudo systemctl restart fh6parse-kiosk.
No 3D cube next to files	Read the chip next to the cube legend: searching... (looking for .stp), rendering... (drawing a hit — wait), no STEP (no matching .stp on stick or model_roots), Z: off (company share not mounted), no CAD (pip3 install -e '[models]'). Rev mismatch counts as no STEP. Print still works.
Cube shows, ticket has no picture	Status not device: /dev/usb/lp0 (CUPS intercepted). Printer rejected GS v 0. Test text-only first (§3.5).
Pictures vanished after --update	--update reinstalls [models] when CAD was already importable. If cubes are still gone, run sudo pip3 install -e '[models]' --break-system-packages.
Pictures still shaded / grey mush	Old cache. rm -rf /tmp/fh6parse-models and wait for the cube again. The current renderer is black edges on white only.
No WARNING: for a wrong D	Clone is older than this pull. git log -1 --oneline must mention D vs T. D is checked on G43 and on G41/G42.
No WARNING: after tapping / G95	Next Txx M6 (or M30) must still be in G95. A G94 on the next tool's line cancels it.
No empty-pocket WARNING: on an idle T	The last Txx M6 with no motion is spindle prep (quiet). Earlier idle Txx M6 should warn.

CLI without the kiosk (reports next to the NC file):

```
python3 -m fh6parse /path/program.nc
python3 -m fh6parse --format 80mm-load --stdout /path/program.nc
python3 -m fh6parse --format 80mm-set --stdout /path/program.nc
python3 -m fh6parse --format 80mm --stdout /path/program.nc
```

(80mm-min is still accepted and prints LOAD. 80mm is RUN.)

7. Files on disk

Path	Role
/home/kiosk/fh6parse	Source checkout (shop update path)
/etc/fh6parse-kiosk.ini	Pins, printer, idle, model_roots (never overwritten by UPDATE)
/home/kiosk/.config/fh6parse/ui.ini	Screen language, mill (including Add mill...), GPIO pins, encoder ticks. Written by settings. Survives UPDATE .
/etc/systemd/system/fh6parse-kiosk.service	Autostart
/etc/sudoers.d/fh6parse-kiosk	NOPASSWD restart for on-screen UPDATE
packaging/fh6parse-kiosk.ini.example	Template

Path	Role
packaging/fh6parse-kiosk.service	Template
/tmp/fh6parse-models	Cached STEP bitmaps (safe to delete)

8. Updating the kiosk

This section is for **fh6parse 1.4.0** on a **git checkout** (/home/kiosk/fh6parse). The package number stays **1.4.0** until you tag a newer release; the kiosk chip, office GUI, and `--version` also show the git short SHA (v1.4.0+abc1234) so you can see which commit is running. Confirm first:

```
python3 -m fh6parse --version
```

You want fh6parse 1.4.0+..... (also **v1.4.0+.....** on the kiosk). The kiosk **never updates by itself**. Print works with or without a network.

8.1 First pull to 1.4.0 (already installed, older number)

If `--version` is older than 1.4.0, SSH as kiosk:

```
cd /home/kiosk/fh6parse
git fetch
git pull --ff-only
python3 -m fh6parse --version
```

Add the sudoers line from **§3.2** if it is missing, then:

```
sudo systemctl restart fh6parse-kiosk
```

From this restart onward, later upgrades use **§8.2**.

If `git pull --ff-only` fails, the clone has local edits or a diverged branch. `git status`. Do not merge on the shop floor.

8.2 On-screen UPDATE (1.3.2 and later)

On each kiosk start, and again when the screensaver wakes (encoder, USB insert, HID, optional spare), a background thread runs `git fetch` (~20 s timeout) and compares HEAD to the tracked branch (@{upstream}, else origin/HEAD, else origin/master / origin/main). A second fetch is skipped while one is already running, while **UPDATE** is already on screen, or while an install is in progress. **Nothing is installed until you tap the button.**

After the check	What you see
Offline, timeout, one-file binary, or already current	No button. v... stays at the top. Print as usual.
Origin has a newer commit	Yellow UPDATE to x.y.z (or a short git hash if the number did not change). Status: v1.4.0 → x.y.z · tap UPDATE to install and restart.

Tap **UPDATE** once (touch or **U**). That is the only action: `git pull --ff-only`, `pip` only if `pyproject.toml` changed, then **restart** fh6parse-kiosk. Print is paused only while that runs. The new version is live after the restart.

kiosk cannot restart a system unit unless you installed the sudoers file in **§3.2**. Without it the pull may succeed and the screen stays on the old process — then:

```
sudo systemctl restart fh6parse-kiosk
```

The check also runs after idle wake, so a commit pushed while the panel was black can show **UPDATE** without a reboot. If the kiosk stayed awake the whole time, restart it (or wait until the next power-on / next sleep-wake) before the button can appear.

8.3 Keyboard / SSH (--update)

Wake the screen if it is black. **Esc** once leaves fullscreen, **Esc** again closes the window if you need a desktop terminal. If systemd owns the display, open a terminal on `:0` or SSH as `kiosk`.

```
python3 -m fh6parse --version
python3 -m fh6parse --update
python3 -m fh6parse --version
```

--update and the on-screen button do this, in order:

- 1. `git pull --ff-only` in the clone (refuses messy merges)
- 2. `pip3 install -e .` **only if** `pyproject.toml` changed; otherwise skips pip. If STEP cubes already work (`[models]` importable), that pip uses `[models]` so pictures stay. A kiosk that never had CAD stays `-e .` and does not pull numpy/trimesh. If cubes still vanish, see **§3.7**.
- 3. `systemctl restart fh6parse-kiosk` if that unit exists; if that fails, `sudo -n systemctl restart fh6parse-kiosk`. Otherwise it prints “restart the kiosk yourself”

It never writes `/etc/fh6parse-kiosk.ini` OR `~/.config/fh6parse/ui.ini`. Pins, printer, idle, `model_roots`, and language stay as you set them.

8.4 One-file ARM tarball

No **UPDATE** button. `python3 -m fh6parse --update` (OR `./fh6parse --update`) exits with a message to copy a new tarball or switch to a git clone. That package is a first copy, not the 1.4.0 upgrade path. To convert: follow **§3.2** (git + pip), point systemd ExecStart back to `python3 -m fh6parse --kiosk --config /etc/fh6parse-kiosk.ini`, then **§8.2**.

8.5 Office GUI (Windows exe / Linux git)

The office window (`fh6parse.exe`, OR `python3 -m fh6parse --gui`) uses the **same one-click UPDATE** as the kiosk: a yellow **UPDATE to x.y.z** (or a git hash) bar under the mill/print row. Nothing is installed until you click it; then the window restarts.

Install	What the bar checks	What one click does
Frozen Windows exe	GitHub latest release asset <code>fh6parse-*_windows-x64.exe</code> newer than 1.4.0	Download, swap the exe, restart
Git checkout (Windows or Linux)	<code>git fetch</code> vs origin, same as §8.2	<code>git pull --ff-only</code> , pip if <code>pyproject.toml</code> changed (keeps <code>[models]</code> if CAD is already there), restart the GUI

The check runs at launch and again when you click back into the window. Offline or already current: no bar. Frozen **1.4.0** PCs only show the bar after you tag a **newer** version (`vX.Y.Z` must match `_version.py`). Git checkouts show a hash while the badge stays **1.4.0**.

9. Field test

Take this sheet to the Pi. The kiosk must show **v1.4.0** at the top right.

1. Get this code onto the Pi (user kiosk):

```
cd /home/kiosk/fh6parse
git fetch
git checkout master
git pull --ff-only
git log -1 --oneline
sudo systemctl restart fh6parse-kiosk
```

git log -1 on current master should mention **mill encoder / LOAD / SET** (or **G54** / travel if that is an older pull). The on-screen badge is still **v1.4.0**. STEP isometrics, D vs T, USB /dev/usb/lp0, cycle time, and kiosk preview are already in older 1.4.0 commits. Or wait for the yellow **UPDATE** after a restart, or after the screen wakes from idle.

Clear old STEP bitmaps:

```
rm -rf /tmp/fh6parse-models
```

If /etc/fh6parse-kiosk.ini still has printer_queue = munbyn, you can leave it (USB is tried first) or comment it out. User kiosk must be in group lp. If the printer is busy:

```
sudo systemctl disable --now cups
```

2. Printer (P047)

Check	Pass
ls -l /dev/usb/lp0 exists, writable by kiosk	
Python test in §3.5 prints TEST and cuts	
RUN ticket status: device:/dev/usb/lp0 (not lp:...)	
Ticket text is readable (PC852 Polish comments), then cut	
Open cover → status Pokrywa otwarta ; RUN / LOAD / SET do not print	
Paper out → status Brak papieru ; RUN / LOAD / SET do not print	
Cover closed, paper in → LOAD (short), SET (offset/cycle), RUN (full) print and cut	
No CUPS garbage / doubled jobs	

3. D and H vs tool number

Use a program with a wrong offset, or a known sample (000814086.nc T10 with H25 D25).

Check	Pass
Wrong H on G43 → WARNING: H... does not match T...	
Wrong D on G43 → WARNING: D... does not match T...	
Wrong D on G41/G42 → same D warning, once	
Matching D (= T) → no D warning	
G95 then next T without G94 → WARNING: G95 still active...	
Idle Txx M6 (not the last change) → empty-pocket warning	
Last Txx M6 with no motion stays quiet (spindle prep)	

Check	Pass
T12 then M6 on the next line counts as one T12 change	
G68 then G69 → G68 T... L... to G69 T... L... on LOAD / SET / RUN	
G68 with no G69 → G68 T... L... without G69	
(...!...) comments listed as programmer notes	
Highlight preview: ops, cycle time, 3D ready, stacked isometric when ready	

4. STEP views (.stp on the USB stick, or model_roots; CAD extra required)

Check	Pass
Stick has Program.nc + matching .stp (root or subfolder)	
3D cube appears when the model is ready (no NAS needed)	
Stacked isometric on the kiosk preview / Windows GUI (not paper only)	
Two stacked views, part not a flat 45° slab (true isometric)	
Black lines on white — no grey shading	
Through-holes as ellipses, not filled blobs	
Long shaft is a thin strip across 80 mm	
Print without a cube is still text-only, no wait	
Settings (F2 / PL chip): language, mill / Add mill... , LOAD	SET
Optional spare GPIO (unwired OK): press sleeps, press wakes; print buttons still ignored while black	
Optional J2 power: omit it and USB-C still boots; do not set WAIT_FOR_POWER_BUTTON without J2	
NVMe Base Duo under the Pi, official 512 GB 2230 in slot A: 1sb1k shows nvme0n1; git checkout lives on that disk	
Panel USB 3.0 (27 mm): stick in the enclosure socket lists .nc; P047 still on a separate Pi USB-A	
Official Active Cooler on FAN header; GPIO screw terminal on the header or a 40-pin riser if it will not fit	
USB-C and HDMI0 clamped to the case; no USB-C extension on the 27 W PSU	
PETG case, vents above the cooler and below the Duo; PCIe flex not creased	
File knob: rest is stable; highlight changes halfway to the next tooth	
Mill knob: mill name next to v... changes; persists like Add mill...	
Colour chips on the bottom of the screen: green LOAD, yellow SET, red RUN left→right	
LOAD has T / H / D / S / Min Z / load boxes, no STEP / offset / chart	
SET has mill, offset, cycle, ! notes, no share chart / each Txx M6 / M00	
RUN has STEP (when ready), offset, time split (G0/F/ATC), tool list, each change, no load boxes / sign-off	

Windows office PC: double-click the exe (or `python -m fh6parse --gui` from a git clone). **Polski / English** radios at the top right (default English). Set **STEP folders.... Add mill...** next to the mill combo (name, rapids m/min, B/C rapid, tool-change seconds, optional max rpm, optional G53 ATC X/Y/Z, work offset X/Y/Z, travel min/max per axis). Next to the preview, tick which **report sections** to include (file/O, mill, printed-at, header comments, ! notes, STEP, offset corners, Ømax/Z, cycle, share chart, **time split**, tool list, load boxes, each Txx M6, M00, extra warnings, sign-off). **Use LOAD pack / SET / RUN** copies a factory pack into those ticks. Preview, Print A4, Print 80 mm, and Save all use the same ticks. There are no LOAD / SET / RUN print buttons on the GUI — those exist only on the kiosk. G68, D vs T, empty pocket, S max, late offset, G95, and travel-too-big always print. Last NC folder, report folder, A4 vs 80 mm, and the section checklist are remembered. If CAM overwrites an open .nc (mtime or size), the preview reloads from disk — wait for **Reading... / Reloaded ...** before print. A wireframe cube means the STEP bitmap is ready; the stacked isometric also appears above the report preview. With mill travel set, the ticket includes the work-

offset origin rectangle in G53 mm when that box is ticked. Windows print is still the browser dialog, not /dev/usb/lp0. If GitHub (frozen exe) or origin (git) has a newer build, a yellow **UPDATE to ...** bar appears under the mill/print row — one click, then the window restarts (same as the kiosk). The check also runs when you click back into the window. Frozen **1.4.0** office boxes only show UPDATE after you tag a **newer** version. Publish by tagging **vX.Y.Z** (GitHub Actions builds it) or packaging\build_windows.bat then packaging\publish_windows.ps1. The tag must match _version.py. Do not overwrite fh6parse-kiosk.ini next to the exe.